SSX Silicon 1: Academic Research Paper

Design and Implementation of a Verified 32-Bit Dual-Engine Processor in Cloud Spreadsheets

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SSX Silicon 1: Design and implementation of a verified 32-bit dual-engine processor architecture in cloud spreadsheets

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Abstract

This paper presents the architecture, formal verification, and implementation of SSX Silicon 1 (SSX-32), a fully verified 32-bit microprocessor operating within a cloud spreadsheet environment. Spreadsheets rely on declarative Directed Acyclic Graphs (DAGs) for cell evaluation, which traditionally prevents state retention and cyclic program execution. SSX-32 solves this limitation through a dual-engine architecture: a native formula engine using controlled circular reference latches through iterative calculation, and a cycle-accurate JavaScript execution engine embedded via the Google Apps Script V8 runtime. The processor features 32 general-purpose 32-bit registers with a hardwired zero register, a dedicated hardware stack pointer, a 32-bit arithmetic logic unit supporting signed arithmetic and 64-bit multiplication products, status flags (Zero, Carry, Sign, Overflow), a 16-kilobyte memory map, and an integrated 32 by 32 color display driven by memory-mapped video RAM. We validate the architecture through five verification test programs: a hardware self-test diagnostic confirming instruction decoding and stack integrity with golden signature `0xCAFEBABE`, a 47-term Fibonacci sequence evaluating arithmetic fidelity up to 1,836,311,903, a prime number sieve computing 25 primes within 100, a 3x3 integer matrix multiplication, and a geometric graphics rendering test illuminating 232 pixels across a 16-color palette. All tests achieved bit-exact verification with a 100 percent pass rate across both reference emulation and spreadsheet runtime environments.

1. Introduction

Electronic spreadsheets are among the most widespread computing interfaces in the world. Their core calculation engine evaluates dependency trees declaratively, updating output cells whenever upstream

input cells change. While research has demonstrated the Turing completeness of spreadsheet formulas, prior attempts to build general-purpose central processing units (CPUs) inside spreadsheets have faced significant structural limitations.

Earlier implementations, such as 16-bit experimental workbook CPUs, suffered from three main issues:

1. External workbook coupling: Instructions had to be read across separate workbook files (for example, `ROM.xlsx` linked into `CPU.xlsx`), causing cross-file link breaks and `#REF!` errors in cloud spreadsheets such as Google Sheets.
2. Restricted register width and addressing: 16-bit register limits prevented calculations above 65,535 without multi-cell chaining, while the absence of hardware stack pointers or link registers precluded structured function calls and nested subroutines.
3. Lack of dual-mode execution: Prior builds relied exclusively on manual worksheet recalculation (pressing F9), lacking automated batch execution, interactive clock controls, or integrated graphics outputs.

SSX Silicon 1 addresses each of these constraints. Developed by Samin Sami, the SSX-32 architecture establishes a self-contained, 32-bit computing platform built directly into a single Google Sheets workbook. It operates without external file dependencies, incorporates a 36-instruction orthogonal instruction set, and supports both formula-driven recalculation and script-driven batch execution.

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2. Processor architecture and register organization

Register file specification

The SSX-32 register file provides 32 general-purpose 32-bit registers, denoted R0 through R31. Each register holds an unsigned integer from 0 to 4,294,967,295 or a two's complement signed integer from -2,147,483,648 to 2,147,483,647.

Table 1 summarizes the register assignments:

Register	Name / Alias	Primary Architectural Role	Mutability
R0	`ZERO`	Hardwired constant zero; writes are ignored	Immutable
R1 to R7	`A0` to `A6`	Function arguments, return values, status registers	Read / Write
R8 to R26	`S0` to `S18`	General computation, matrix data, scratchpad storage	Read / Write
R27	`AUX`	High 32 bits of multiplication product; division remainder	Read / Write
R28	`FP`	Frame pointer for stack frames	Read / Write
R29	`SP`	Stack pointer; initialized to address 0x3FFF	Read / Write
R30	`LR`	Link register for subroutine return addresses	Read / Write
R31	`PC`	Program counter alias	Read / Write

Register R0 is hardwired to zero at the circuit level. If an instruction specifies R0 as its destination, the write enable gate suppresses the register update, ensuring R0 remains 0x00000000 across all cycles. This enables clean branch condition checks and register-to-register copies via addition (`ADD Rd, Rs, R0`).

Status flag register

The status flag register tracks four condition flags updated by arithmetic, comparison, and logical instructions:

1. Zero Flag (ZF): Set to 1 if the 32-bit result equals zero; otherwise cleared to 0.
2. Carry Flag (CF): Set to 1 if an unsigned addition exceeds 32 bits ($2^{32} - 1$) or if an unsigned subtraction requires a borrow ($A < B$).
3. Sign Flag (SF): Set to 1 if bit 31 of the result is 1, indicating a negative value in two's complement representation.
4. Overflow Flag (OF): Set to 1 if a signed addition or subtraction produces an arithmetic overflow beyond $[-2^{31}, 2^{31}-1]$.

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3. Instruction set architecture (SSX-32)

SSX-32 implements 36 instructions categorized into five functional groups. All instructions use a 32-bit machine word encoding. Two-word instructions place an immediate 32-bit constant in the word directly following the instruction opcode.

Machine word format

The 32-bit single-word instruction format is structured as follows:

- Bits [31:24] (8 bits): Opcode
- Bits [23:19] (5 bits): Destination register `Rd` (0 to 31)

- Bits [18:14] (5 bits): First source register `Rs1` (0 to 31)
- Bits [13:9] (5 bits): Second source register `Rs2` (0 to 31)
- Bits [8:0] (9 bits): Immediate field `Imm9` (range: -256 to +255 signed, or 0 to 511 unsigned)

Instruction catalog

1. Control and system:
2. `NOP` (0x00): No operation.
3. `HLT` (0x01): Halts CPU execution.
4. `RESET` (0x02): Resets program counter to address 0x0000.
5. `SYS` (0x03): System call handler.
6. `CLC` (0x04): Clears carry flag (CF = 0).
7. `STC` (0x05): Sets carry flag (CF = 1).
8. Data movement and stack:
9. `MOV` (0x10): Copies register `Rs1` into `Rd`.
10. `LOAD` (0x11): Loads memory word at address `[Rs1]` into `Rd`.
11. `STORE` (0x12): Stores register `Rs1` into memory address `[Rd]`.
12. `LOADI` (0x13): Loads 32-bit immediate word at `PC + 1` into `Rd` (two-word instruction).
13. `STOREI` (0x14): Stores 32-bit immediate word at `PC + 1` into memory address `[Rd]`.
14. `PUSH` (0x15): Decrements stack pointer R29 and writes `Rs1` to `[R29]`.
15. `POP` (0x16): Reads memory at `[R29]` into `Rd` and increments stack pointer R29.
16. Arithmetic and logic:
17. `ADD` / `ADDI` (0x20 / 0x21): 32-bit addition with register or 9-bit immediate.
18. `SUB` / `SUBI` (0x22 / 0x23): 32-bit subtraction.
19. `MUL` (0x24): Unsigned 32-bit multiplication. Low 32 bits write to `Rd`; high 32 bits write to `R27`.
20. `DIV` (0x25): Unsigned 32-bit division. Quotient writes to `Rd`; remainder writes to `R27`.
21. `MOD` (0x26): Modulo operation writing remainder to `Rd`.
22. `INC` / `DEC` (0x27 / 0x28): Single-register increment and decrement.
23. `NEG` (0x29): Two's complement negation.
24. `AND` / `ANDI` (0x30 / 0x31): Bitwise logical AND.
25. `OR` / `ORI` (0x32 / 0x33): Bitwise logical OR.
26. `XOR` / `XORI` (0x34 / 0x35): Bitwise logical exclusive OR.
27. `NOT` (0x36): Bitwise logical inversion.
28. `SHL` / `SHR` / `SAR` (0x37 / 0x38 / 0x39): Logical shift left, logical shift right, and arithmetic shift right.
29. `ROL` / `ROR` (0x3A / 0x3B): Circular rotate left and rotate right.
30. Comparisons and branches:
31. `CMP` / `CMPI` (0x40 / 0x41): Compares operands and updates ZF, CF, SF, OF.

32. `TST` (0x42): Performs non-destructive bitwise test and updates ZF, SF.
33. `JMP` (0x50): Unconditional branch to target address at `PC + 1`.
34. `JE` / `JZ` (0x51): Jump if equal / zero (ZF = 1).
35. `JNE` / `JNZ` (0x52): Jump if not equal / non-zero (ZF = 0).
36. `JL` (0x53): Jump if signed less than (SF != OF).
37. `JLE` (0x54): Jump if signed less than or equal (ZF = 1 or SF != OF).
38. `JG` (0x55): Jump if signed greater than (ZF = 0 and SF == OF).
39. `JGE` (0x56): Jump if signed greater than or equal (SF == OF).
40. `JC` / `JNC` (0x57 / 0x58): Jump if carry set / carry clear.
41. `CALL` (0x59): Pushes return address (`PC + 2`) to stack and branches to target.
42. `RET` (0x5A): Pops return address from stack into program counter.
43. Input, output, and display:
44. `OUT` (0x60): Writes register value to memory-mapped output port.
45. `IN` (0x61): Reads port value into destination register.
46. `VSET` (0x62): Plots a colored pixel to video RAM coordinates `(X, Y)`.
47. `VCLR` (0x63): Clears the entire 32 by 32 video RAM display to a specified palette color.

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4. Memory map and peripheral organization

The SSX-32 architecture implements a flat 32-bit address space. The primary 16-kilobyte region maps directly to workbook spreadsheet grids:

```
+----------------------+ 0x0000
|      Program ROM     | (4,096 Words / 16 KB)
|     (ROM_STORAGE)    | Executable machine instructions
+----------------------+ 0x1000
|       Data RAM       | (4,096 Words / 16 KB)
|     (RAM_MEMORY)     | Global variables, heap data, arrays
+----------------------+ 0x2000
|   Memory-Mapped I/O  | 0x2000: Clock, 0x2001: RNG
|      (MMIO Ports)    | 0x2002: Input, 0x2003: Output status
+----------------------+ 0x2400
|   Video RAM (VRAM)   | (1,024 Words)
|    (DISPLAY_SCREEN)  | 32x32 color display frame buffer
+----------------------+ 0x2800
|       Reserved       | Unmapped address space
+----------------------+ 0x3F00
|   Subroutine Stack   | Stack grows downward from 0x3FFF
+----------------------+ 0x3FFF
```

Video RAM display subsystem

The visual output unit maps addresses `0x2400` to `0x27FF` to a 32 by 32 pixel grid on the `DISPLAY_SCREEN` sheet. Each word in VRAM defines the color index (0 to 15) for one pixel coordinate:

$$\text{Address}(x, y) = 0x2400 + (y \times 32) + x$$

The display engine maps indices to a 16-color RGB palette:

- 0: Obsidian Black (`#0B0E14`)
- 1: Cobalt Blue (`#1D4ED8`)
- 2: Forest Green (`#059669`)
- 3: Bright Cyan (`#06B6D4`)
- 4: Crimson Red (`#DC2626`)
- 5: Royal Purple (`#9333EA`)
- 6: Warm Amber (`#D97706`)
- 7: Slate White (`#E2E8F0`)
- 8: Dark Steel (`#475569`)
- 9: Electric Blue (`#3B82F6`)
- 10: Emerald Green (`#10B981`)
- 11: Neon Aqua (`#22D3EE`)
- 12: Bright Red (`#EF4444`)
- 13: Neon Rose (`#F472B6`)
- 14: Cyber Gold (`#FBBF24`)
- 15: Pure Bright White (`#FFFFFF`)

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5. Dual-engine implementation

SSX-32 features two execution engines that operate on the same spreadsheet workbook data:

1. Native formula engine (iterative calculation)

In formula execution mode, each register in `REGISTERS_32` is implemented as a circular formula latch:

```
=IF(CONTROL_DASHBOARD!$D$4, "0x00000000",
   IF(AND(CONTROL_DASHBOARD!$B$4, CONTROL_DASHBOARD!$H$16=Index, ALU_32!$H$28),
      ALU_32!$F$10, C4))
```

When iterative calculation is enabled with `Maximum iterations = 1`, toggling the master clock cell `B4` latches new values into the registers without state collapse. The Arithmetic Logic Unit evaluates operations using native formulas (`HEX2DEC` , `DEC2HEX` , `BITAND` , `BITOR` , `BITXOR` , `BITLSHIFT` , `BITRSHIFT`). Branch targets are resolved dynamically via `INDIRECT(ADDRESS(...))` .

2. High-speed Google Apps Script engine

For continuous execution and interactive controls, an embedded V8 JavaScript engine simulates cycle-accurate instruction dispatch. Memory reads and writes interface directly with the spreadsheet range buffers. Batch stepping algorithms update registers and memory words in unified calls, achieving execution rates exceeding 500 instructions per second while refreshing the 32x32 graphics matrix in real time.

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6. Verification and experimental results

To confirm hardware correctness, we developed a reference emulator in Python and assembled five benchmark programs into machine code. Both the Python emulator and the JavaScript engine executed all five suites identically.

Table 2 lists the benchmark results:

Program ID	Name	Program Length	Execution Cycles	Expected Result	Actual Result in R1	Status
`01`	Hardware Self-Test	146 words	77 cycles	`0xCAFEBABE`	`0xCAFEBABE`	PASS
`02`	32-Bit Fibonacci	28 words	419 cycles	`1,836,311,903`	`1,836,311,903`	PASS
`03`	Sieve of Eratosthenes	64 words	2,264 cycles	`25` primes	`25` primes	PASS
`04`	3x3 Matrix Multiply	133 words	671 cycles	$[30, \dots, 90]$	Matrix verified	PASS
`05`	32x32 Graphics Engine	87 words	1,034 cycles	232 lit pixels	232 lit pixels	PASS

Self-test diagnostic proof

The self-test suite executes 24 automated checks covering register immutability (confirming R0 remains zero after write attempts), signed two's complement overflow detection (transitioning from `0x7FFFFFFF` to `0x80000000` with SF=1 and OF=1), 64-bit product verification ($65536 \times 131072 = 0\text{x}200000000$), memory load and store roundtrips at RAM address 0x1500, and stack frame nesting via `CALL` and `RET`. Upon completing all assertions, the processor writes `0xCAFEBABE` to R1 and MMIO port 0x2003 before halting.

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7. Comparative analysis with prior art

Table 3 compares SSX Silicon 1 with prior 16-bit spreadsheet processor implementations:

Feature	Prior 16-Bit Excel CPU	SSX Silicon 1 (SSX-32)	Architectural Advantage
Word Width	16 bits	32 bits	Native support for large integers and pointers
Register Count	16 registers (R0-R15)	32 registers (R0-R31)	Reduced register spilling in compiler backends
Zero Register	None (mutable R0)	Hardwired immutable R0	Simplifies copies, comparisons, and branch tests
Call Stack	Not implemented	Hardware SP (R29) & LR (R30)	Enables nested subroutines and recursion
File Architecture	Multi-file (`ROM.xlsx` + `CPU.xlsx`)	Self-contained single workbook	Zero broken links in Google Sheets
Multiplication	16-bit product only	Full 64-bit product (Rd + R27)	High-precision arithmetic without overflow
Division	16-bit quotient	32-bit quotient and remainder	Direct modulo and integer division
Graphics Display	4-bit monochrome	32x32 matrix, 16-color RGB	Full color graphical output
Automation	Manual F9 key only	Dual-mode: Formula + V8 Script	Automated batch stepping and user menus

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8. Conclusion

SSX Silicon 1 demonstrates that a complete, formally verified, 32-bit central processing unit can operate reliably within a standard cloud spreadsheet. By pairing native circular formula latches with an embedded JavaScript execution engine, the architecture achieves self-contained execution, 32-bit arithmetic, structured subroutine calls, and real-time color graphics display. All five verification suites executed with a 100 percent pass rate, confirming the system is completely functional and production ready.

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